

VoteSecure - Full Source Code Documentation

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1. Backend - server.js

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
require("dotenv").config();

const app = express();
app.use(express.json());
app.use(cors());

mongoose.connect("mongodb://127.0.0.1:27017/votingDB")
  .then(()=>console.log("DB Connected"))
  .catch(err=>console.log(err));

app.use("/api/auth", require("./routes/auth"));
app.use("/api/vote", require("./routes/vote"));
app.use("/api/admin", require("./routes/admin"));

app.listen(5000, ()=>console.log("Server running on port 5000"));
```

models/User.js

```
const mongoose = require("mongoose");

const userSchema = new mongoose.Schema({
  name: String,
  email: {type:String, unique:true},
  password: String,
  role: {type:String, default:"user"},
  hasVoted: {type:Boolean, default:false}
});

module.exports = mongoose.model("User", userSchema);
```

models/Candidate.js

```
const mongoose = require("mongoose");

const candidateSchema = new mongoose.Schema({
  name: String,
  votes: {type:Number, default:0}
});

module.exports = mongoose.model("Candidate", candidateSchema);
```

middleware/auth.js

```
const jwt = require("jsonwebtoken");

module.exports = function(req,res,next){
  const token = req.header("Authorization");
  if(!token) return res.status(401).json("No token");

  try{
    const decoded = jwt.verify(token,"secretkey");
    req.user = decoded;
    next();
  }catch{
    res.status(401).json("Invalid token");
  }
};
```

routes/auth.js

```
const router = require("express").Router();
const User = require("../models/User");
const bcrypt = require("bcryptjs");
const jwt = require("jsonwebtoken");

router.post("/register", async (req,res)=>{
  const {name,email,password} = req.body;
  const hashed = await bcrypt.hash(password,10);
  const user = new User({name,email,password:hashed});
  await user.save();
  res.json("Registered");
});

router.post("/login", async (req,res)=>{
  const {email,password} = req.body;
  const user = await User.findOne({email});
  if(!user) return res.status(400).json("User not found");

  const match = await bcrypt.compare(password,user.password);
  if(!match) return res.status(400).json("Wrong password");

  const token = jwt.sign({id:user._id, role:user.role},"secretkey");
  res.json({token});
});

module.exports = router;
```

routes/vote.js

```
const router = require("express").Router();
const Candidate = require("../models/Candidate");
const User = require("../models/User");
const auth = require("../middleware/auth");

router.get("/", async (req,res)=>{
  const candidates = await Candidate.find();
  res.json(candidates);
});

router.post("/:id", auth, async (req,res)=>{
  const user = await User.findById(req.user.id);
  if(user.hasVoted) return res.json("Already voted");

  const candidate = await Candidate.findById(req.params.id);
  candidate.votes += 1;
  await candidate.save();

  user.hasVoted = true;
  await user.save();

  res.json("Vote counted");
});

module.exports = router;
```

routes/admin.js

```
const router = require("express").Router();
const Candidate = require("../models/Candidate");

router.post("/add", async (req,res)=>{
  const {name} = req.body;
  const c = new Candidate({name});
  await c.save();
  res.json("Added");
});
```

```
router.get("/results", async (req,res)=>{
  const data = await Candidate.find();
  res.json(data);
});

module.exports = router;
```

2. Frontend - App.js

```
import {BrowserRouter,Routes,Route} from "react-router-dom";
import Login from "./Login";
import Register from "./Register";
import Vote from "./Vote";
import Admin from "./Admin";

function App(){
  return(
    <BrowserRouter>
      <Routes>
        <Route path="/" element={<Login/>}/>
        <Route path="/register" element={<Register/>}/>
        <Route path="/vote" element={<Vote/>}/>
        <Route path="/admin" element={<Admin/>}/>
      </Routes>
    </BrowserRouter>
  );
}
export default App;
```

Login.js

```
import {useState} from "react";
import axios from "axios";

export default function Login(){
  const [data,setData] = useState({email:"",password:""});

  const login = async ()=>{
    const res = await axios.post("http://localhost:5000/api/auth/login",data);
    localStorage.setItem("token",res.data.token);
    window.location="/vote";
  };

  return(
    <div>
      <input placeholder="Email" onChange={e=>setData({...data,email:e.target.value})}/>
      <input type="password" placeholder="Password" onChange={e=>setData({...data,password:e.target.value})}/>
      <button onClick={login}>Login</button>
    </div>
  );
}
```

Register.js

```
import {useState} from "react";
import axios from "axios";

export default function Register(){
  const [data,setData] = useState({name:"",email:"",password:""});

  const register = async ()=>{
    await axios.post("http://localhost:5000/api/auth/register",data);
    alert("Registered");
  };

  return(
    <div>
      <input placeholder="Name" onChange={e=>setData({...data,name:e.target.value})}/>
      <input placeholder="Email" onChange={e=>setData({...data,email:e.target.value})}/>
      <input type="password" placeholder="Password" onChange={e=>setData({...data,password:e.target.value})}/>
      <button onClick={register}>Register</button>
    </div>
  );
}
```

Vote.js

```
import {useEffect,useState} from "react";
import axios from "axios";

export default function Vote(){
  const [candidates,setCandidates] = useState([]);

  useEffect(()=>{
    axios.get("http://localhost:5000/api/vote")
      .then(res=>setCandidates(res.data));
  },[]);

  const vote = async(id)=>{
    const token = localStorage.getItem("token");

    await axios.post(`http://localhost:5000/api/vote/${id}`, {},
      {headers:{Authorization:token}});

    alert("Voted!");
  };

  return(
    <div>
      {candidates.map(c=>(
        <div key={c._id}>
          {c.name}
          <button onClick={()=>vote(c._id)}>Vote</button>
        </div>
      ))}
    </div>
  );
}
```

Admin.js

```
import {useState,useEffect} from "react";
import axios from "axios";

export default function Admin(){
  const [name,setName] = useState("");
  const [candidates,setCandidates] = useState([]);

  const add = async ()=>{
    await axios.post("http://localhost:5000/api/admin/add",{name});
    alert("Added");
  };

  useEffect(()=>{
    axios.get("http://localhost:5000/api/admin/results")
      .then(res=>setCandidates(res.data));
  },[]);

  return(
    <div>
      <input onChange={e=>setName(e.target.value)} placeholder="Candidate"/>
      <button onClick={add}>Add</button>

      {candidates.map(c=>(
        <div key={c._id}>{c.name} - {c.votes}</div>
      ))}
    </div>
  );
}
```